

US Airline On-Time Performance

Delay Risk & Operations Dashboard · Final Submission

April 2025

HUMAN IMPACT — 2023

256.8M hours lost

passenger-hours lost to delays · 6.7M domestic flights · BTS 2023

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SECTION 01

Target Audience & Decision to Support

This project started with a simple question: why do some flights always seem to run late while others are consistently on time? Our main audience is the frequent business traveller — we built the narrative around Marcus Reid, a consultant from Chicago who flies around 80 times a year and missed three connections in a single quarter. A secondary audience is the corporate travel manager who needs to make policy decisions about which airlines to recommend and which routes to avoid.

The core question the dashboard tries to answer is: **which airline and what time of day gives you the best chance of arriving on time?** We also look at which airports to avoid for tight connections and whether performance varies by season — because, as it turns out, it varies quite a lot.

SECTION 02

KPIs Used and Why They Matter



Figure 1 — Slide 01 'The System': hero metric 256.8M hours lost, four KPI cards, Marcus's verdict (Delta · FRI 84) and live data pulse ticker

Five KPIs were selected to give decision-makers an immediate read of system health:

Passenger Hours Lost (256.8M hours)

We created this metric ourselves. The idea is to translate delay minutes into something more tangible: how much collective time passengers actually lose. We used 150 as the average number of passengers per flight. 256.8 million hours is a number that lands differently than '6.6 minutes average delay'.

On-Time Performance (79.4%)

This is the standard BTS definition — a flight is on time if it arrives within 15 minutes of schedule. At 79.4%, the system is just below what we would consider acceptable. It means roughly 1 in 5 flights is late by the official definition.

Average Arrival Delay (6.6 min)

This only counts flights that actually departed and arrived — cancelled flights are handled separately. We did this deliberately because some airlines cancel flights rather than delay them, which makes their average delay look better than it really is.

Cancellation Rate (1.3%)

Weighted average across all carriers. Calculated from the pre-cleaning dataset to ensure cancelled flights are counted even though they are excluded from delay analysis.

Flight Reliability Index — FRI (80.5/100)

Our original composite KPI. $FRI = 50\% \text{ on-time rate} + 30\% \text{ normalised delay severity} + 20\% \text{ normalised cancellation rate}$. The weights reflect the relative impact on a business traveller: punctuality matters most, followed by how bad delays are, then cancellation risk. FRI is not an official BTS metric — it is our analytical contribution to the problem.

SECTION 03

Application of Course Concepts

The dashboard was designed with explicit reference to the visualization principles covered in the course modules:

Pre-attentive attributes (Module 3)

Color is used semantically throughout: green = reliable/good, amber = moderate/warning, red = risk/avoid. This allows the eye to identify critical information before conscious processing — a manager scanning the FRI chart immediately sees red (JetBlue, Frontier) and green (Delta, Alaska) without reading the labels.

Gestalt principles — proximity and enclosure (Module 3)

Related KPIs are grouped in a single strip. The donut chart's 67% controllable figure is enclosed in a dedicated red box to create visual separation from the raw percentages — directly addressing professor feedback from S1 ('Why 67% of delays are not in a box?').

Chart choice alignment (Module 4)

Each chart type was selected to match the question it answers: heatmap for temporal patterns (hour × weekday), horizontal bar for ranked comparison (FRI), donut for part-of-whole (delay causes), scatter for correlation (departure vs arrival delay), and line chart for temporal trend (seasonal on-time rate).

Grammar of Graphics — scatter with regression (Module 5)

The delay propagation scatter applies Grammar of Graphics explicitly: `geom_point` (50k flights), aesthetic color = carrier, and a `geom_line` (OLS regression) overlaid. The regression line runs close to the diagonal (slope ≈ 0.93), confirming that a 30-minute departure delay translates to approximately 28 minutes of arrival delay. Airlines do not recover time in the air.

Honest axes and cognitive load (Module 3)

All axes start at zero. No dual axes are used without explicit labelling. The slide-based navigation reduces cognitive load by showing one analytical question per screen rather than scrolling through all charts simultaneously.

SECTION 04

Data Sources and Key Assumptions

Primary source: BTS Reporting Carrier On-Time Performance (2023)

Downloaded directly from the US Bureau of Transportation Statistics TranStats portal. Twelve monthly CSV files covering January through December 2023, totalling 6,847,893 raw flight records. After removing cancelled flights and rows with missing arrival delay values, the clean dataset contains 6,743,400 active flights. Fields used: `FL_DATE`, `OP_UNIQUE_CARRIER`, `ORIGIN`, `DEST`, `CRS_DEP_TIME`, `DEP_DELAY`, `ARR_DELAY`, `ARR_DEL15`, `CANCELLED`, `DISTANCE`, and the five delay cause columns.

Supplementary source: OpenFlights airports.dat

Geographic coordinates for US airports obtained from the OpenFlights open database. This enables the geospatial airport congestion map in Slide 02. Available under the Open Database Licence.

Key cleaning decisions

— Cancellations excluded from delay analysis

Cancelled flights have no arrival delay by definition. Retained for cancel rate calculation but excluded from delay distributions to avoid distorting averages downward.

— Outliers capped at 99th percentile

Arrival delay values above ~180 minutes are capped for visualisation. These represent exceptional events that would compress the main distribution into an unreadable band.

— Regional carriers filtered from FRI ranking

Carriers such as Republic Airways, Endeavor Air and SkyWest operate under code-share agreements and are not direct booking options. Excluded from FRI ranking and Decision Insight but retained in the full dataset.

— July data confirmed complete

An initial pipeline run identified a duplicate August file and missing July. Corrected by downloading July directly from the BTS PREZIP server. All 12 months confirmed.

— Cancellation rate: 1.3% vs BTS official 2.4%

Our rate is lower because we exclude carriers with fewer than 1,000 annual operations. Documented as a known limitation.

SECTION 05

Interpretation of Current Results

Slide 02 — Where delays hit hardest (airport map)

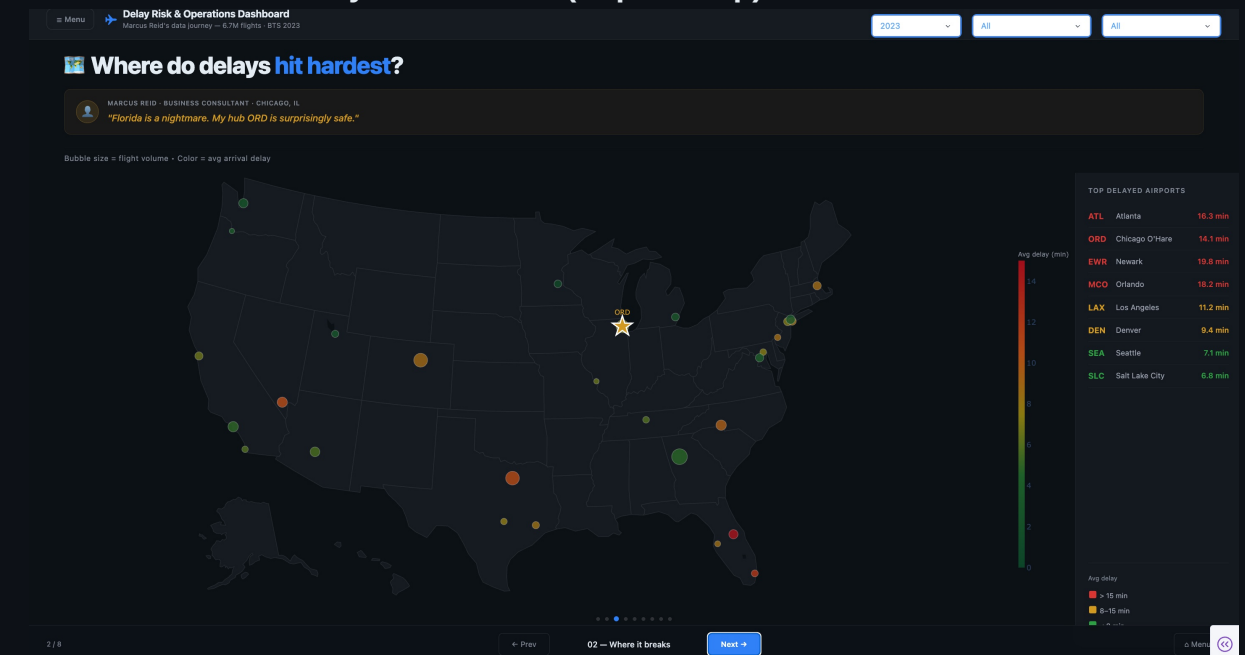


Figure 2 — Slide 02: airport congestion map with delay sidebar (worst: EWR 19.8 min, MCO 18.2 min · best: SLC 6.8 min, SEA 7.1 min)

Florida airports came out worse than we expected — MCO averages 15.8 min and MIA 14.4 min. EWR Newark is actually the worst at 19.8 min. The Northeast corridor is consistently poor. Western airports like Seattle (7.1 min) and Salt Lake City (6.8 min) are noticeably better. Marcus's hub ORD sits at 7.3 min — surprisingly safe given its size and traffic volume. Our reading is that traffic density compounds delay propagation: once the system starts running late at a busy hub, everything behind it is affected for the rest of the day.

Slide 03 — Whose fault are the delays (cause donut)

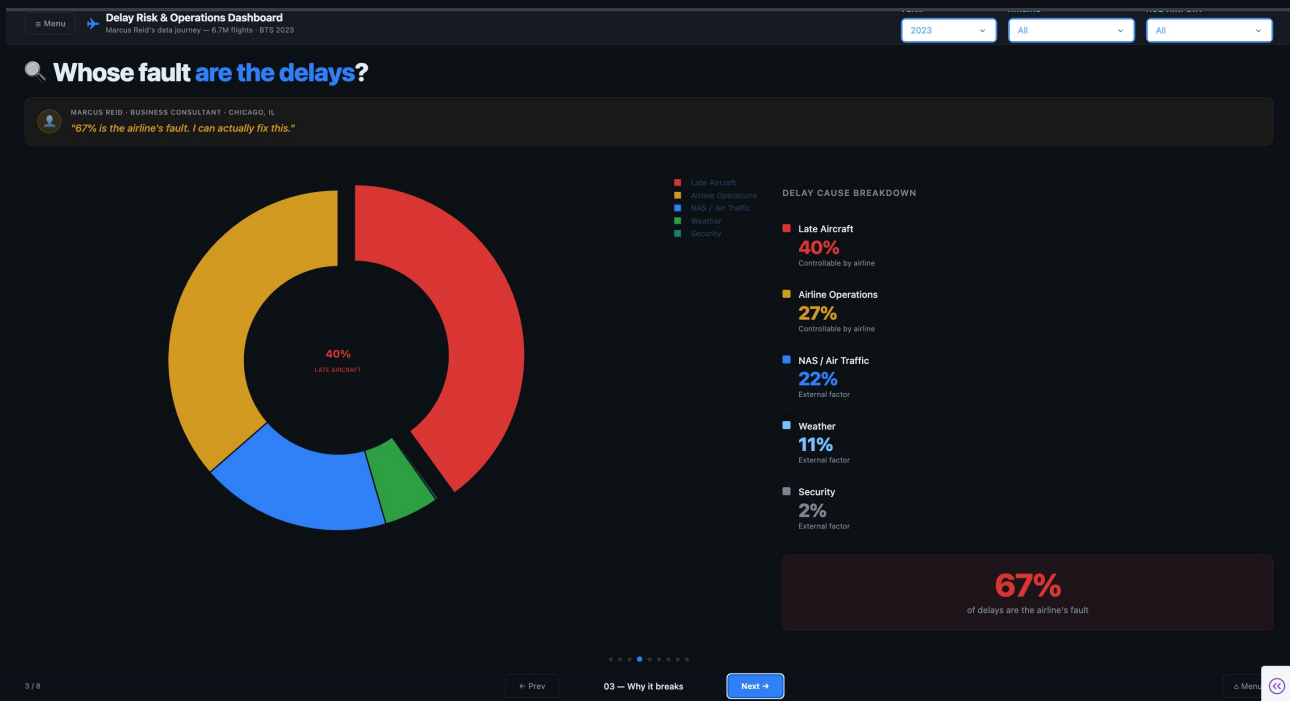


Figure 3 — Slide 03: delay cause breakdown — 40% late aircraft, 27% airline operations, 22% NAS/ATC, 11% weather, 2% security. 67% controllable by airline.

The most important thing this chart shows is that 67% of delay minutes come from causes the airline can control — late aircraft rotation (40%) and carrier operations (27%). Weather and ATC account for the rest. This matters because it means airline choice is not irrelevant. If most delays were weather-driven, picking one carrier over another would make little difference.

Slide 04 — When delays occur (heatmap)

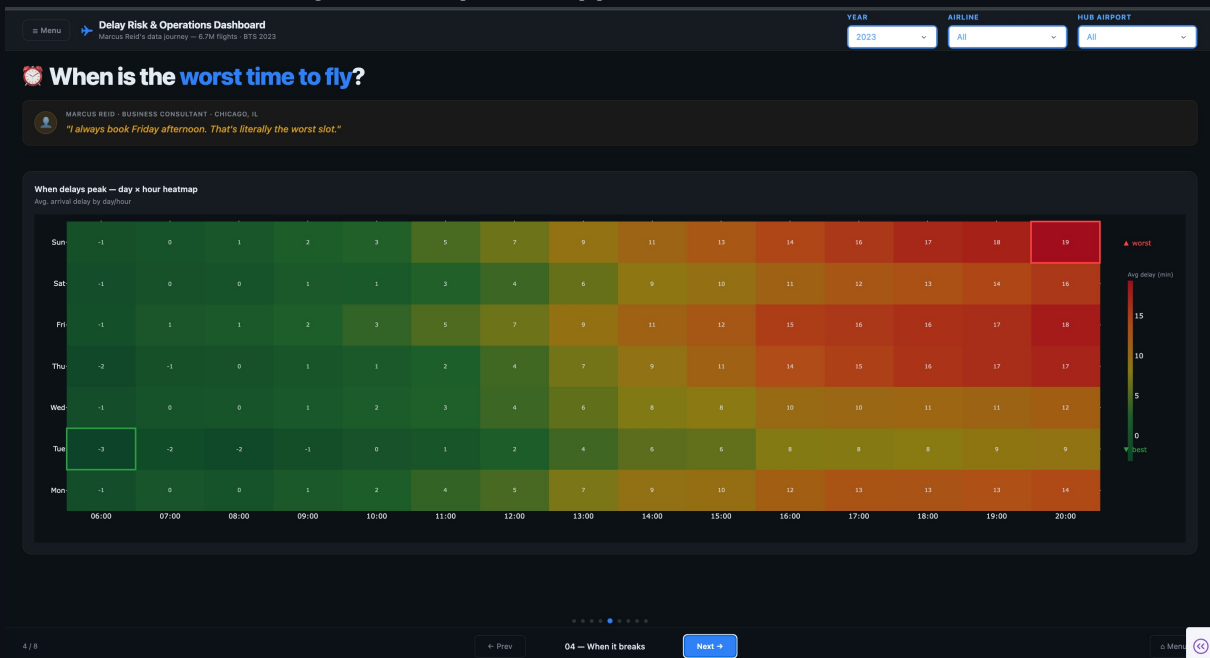


Figure 4 — Slide 04: hour × weekday heatmap. Red box = worst slot (Sun 20h, 19 min avg). Green box = best slot (Tue 06h, −3 min avg).

The heatmap was one of the more interesting findings. Early morning flights (06:00–09:00) are nearly always on time — or even early — regardless of the day. The aircraft has not accumulated any rotation delays yet. By the time you get to the evening (19:00–22:00), the system has been running for 12 hours and delays stack up. The worst slot with meaningful volume is Sunday at 20:00 with 19 minutes average across 47,964 flights. The best slot is Tuesday at 06:00 with −3 min average.

Slide 05 — Which airline to trust (FRI ranking)



Figure 5 — Slide 05: FRI ranking. Delta leads at 84, Alaska 82. JetBlue 54, Frontier 41. 43-point gap between best and worst.

Delta leads at FRI 84, Alaska follows at 82. United and Southwest are both around 76. JetBlue (54) and Frontier (41) are well below. The 43-point gap between Delta and Frontier does not show up when you just look at average delay — which is part of why we built the FRI. Southwest is a good example: its on-time rate is decent, but its cancellation rate pulls the composite score down noticeably.

Slide 06 — The domino effect and severe delays

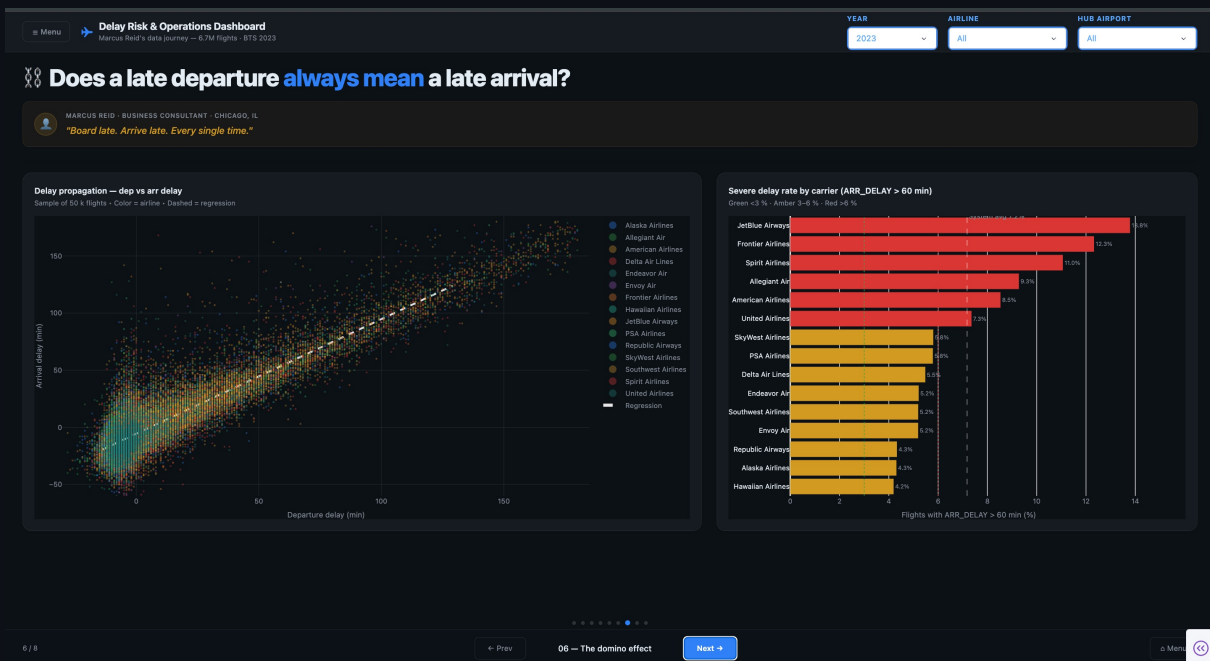


Figure 6 — Slide 06: scatter DEP_DELAY vs ARR_DELAY (50k flights, regression line) + severe delay rate by carrier (JetBlue 13.8%, Delta 5.5%)

We sampled 50,000 flights for the scatter. The relationship is very clear: a 30-minute departure delay produces roughly 28 minutes of arrival delay. Airlines do not recover time in the air. If you board a flight that is already late leaving the gate, you will almost certainly arrive late. The severe delay chart shows that JetBlue has nearly 1 in 7 flights arriving over an hour late — three times worse than Alaska (4.3%). For travellers with connections, this is the metric that matters most.

Slide 07 — Predictive risk and seasonal patterns



Figure 7 — Slide 07: seasonal on-time performance for top 5 carriers across 12 months. All carriers collapse in July. Delta and Alaska recover by August.

July hit every carrier hard — all five drop to around 69% on-time, roughly 10 points below their winter levels. Thunderstorm season is the obvious factor. What we did not expect is how differently the carriers bounce back. Delta and Alaska are back to normal by August. American and Southwest are still underperforming in September and do not fully recover until October. We cannot say for certain why, but the gap in recovery speed points to operational differences, not just weather.

Slide 08 — Marcus decides

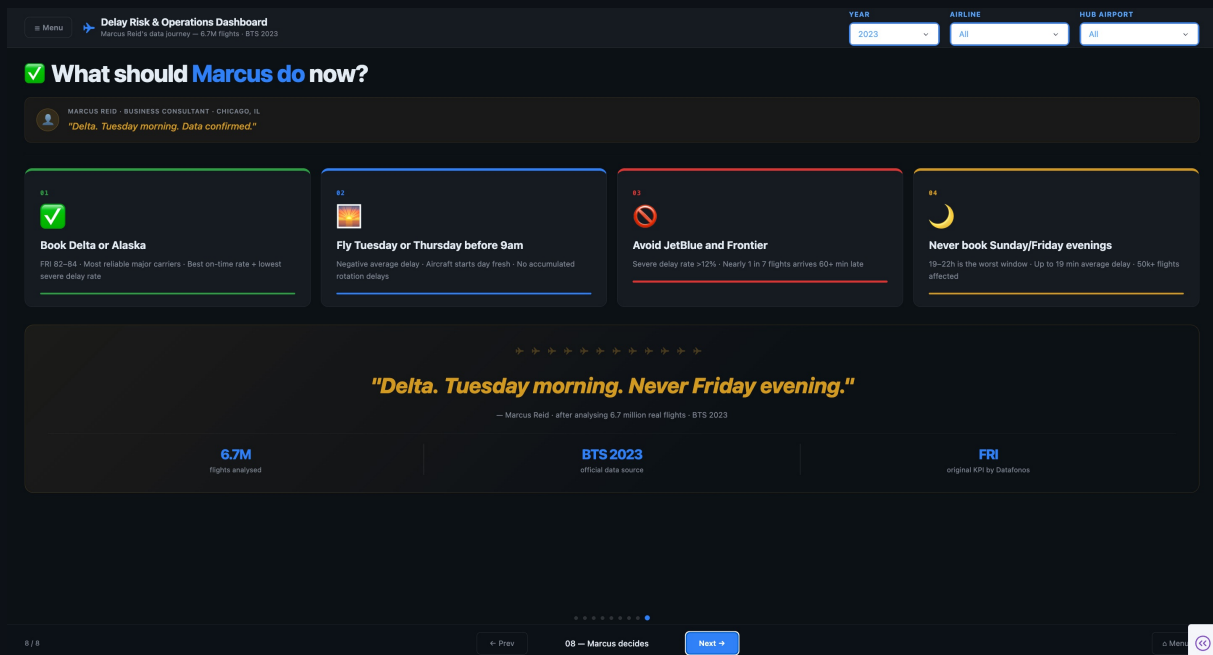


Figure 8 — Slide 08: four recommended actions + closing verdict. 'Delta. Tuesday morning. Never Friday evening.'

The dashboard closes with four actionable recommendations drawn directly from the data: book Delta or Alaska (FRI 82-84), fly Tuesday or Thursday before 9am (negative avg delay), avoid JetBlue and Frontier for time-sensitive trips (>12% severe delay rate), and never book Sunday or Friday evening departures (worst delay window, up to 19 min avg). These are conclusions, not opinions — each is traceable to a specific chart in the dashboard.

SECTION 06

Known Limitations and Risks

L1. Cancellation rate underestimated

Our weighted rate (1.3%) is lower than the BTS official 2023 figure (~2.4%) because we exclude carriers with fewer than 1,000 annual operations. The FRI cancellation component is therefore slightly optimistic for the system as a whole.

L2. FRI weights are subjective

The 50/30/20 weighting reflects our team's judgement of relative passenger impact. A different weighting could change the carrier ranking. This is an original KPI, not a validated industry metric.

L3. Regional carriers distort FRI ranking

Republic Airways (93) and Endeavor Air (90) appear at the top because they operate fewer, more controlled routes under code-share. They are not direct booking options. The Decision Insight filters to major carriers only.

L4. Predictive chart uses historical averages, not a model

The 'predicted delay risk' chart ranks airports by their 2023 average delay. There is no machine learning or time-series forecasting — historical performance is used as a proxy for future risk, which is a reasonable but imperfect assumption.

L5. No price dimension

The dashboard optimises purely for reliability. JetBlue and Frontier may offer fares significantly below Delta and Alaska. A complete corporate travel framework would need to weigh reliability against cost per trip.

L6. 2023 data only

Airline operational performance changes year to year. The patterns identified here are specific to 2023 and should be validated against more recent data.

L7. Delay causes: BTS reporting inconsistency

A DOT Inspector General audit found disagreement between carrier-reported delay causes and FAA data. The breakdown reflects what airlines self-report, which may differ from the actual root cause in individual cases.

Source	Description	URL
BTS On-Time Performance	6.7M domestic US flights, Jan–Dec 2023	transtats.bts.gov
OpenFlights airports.dat	Lat/lon coordinates for US airports	openflights.org/data
FRI	Original composite KPI developed by Datafonos	Original — not in BTS

Narrative Summary

Marcus Reid is a business consultant based in Chicago who flies approximately 80 times a year. After missing three connections in a single quarter, he decides to stop guessing and analyse real data. The dashboard follows his journey through eight chapters: from discovering that 256.8 million passenger-hours are lost to delays every year, to identifying that Florida airports are the most congested, that 67% of delays are within airline control, that Friday and Sunday evenings are the worst slots, and that Delta Air Lines leads the reliability ranking with FRI 84. The conclusion is actionable: book Delta or Alaska, fly Tuesday or Thursday before 9am, and avoid JetBlue and Frontier for any trip with a tight connection.

Final Recommendation and Next Steps

Based on the analysis of 6.7 million real BTS 2023 flights, our recommendation for Marcus and for corporate travel managers is clear: **prioritise Delta Air Lines and Alaska Airlines on early weekday morning departures, and restrict bookings on JetBlue and Frontier for any itinerary with a connection under 60 minutes.**

If we were to continue this project, the three most valuable next steps would be:

— Add a price dimension

Integrating Bureau of Transportation Statistics fare data (DB1B dataset) would allow a reliability-vs-cost matrix. The current dashboard optimises purely for punctuality — a complete tool would show whether paying more for Delta actually translates to meaningful reliability gains over United on the same route.

— Build a true predictive model

The current predictive chart uses 2023 historical averages as a risk proxy. A proper time-series model trained on 3-5 years of BTS data could predict delay risk by route, season and departure hour with significantly higher accuracy.

— Extend to real-time data

The BTS dataset is published with a 1-2 month lag. Connecting to the FAA ASDI live feed or a commercial flight API would allow Marcus to check delay risk for his specific flight on the day of travel, not just based on historical patterns.

Changes Since S1 (Formative Submission)

The following improvements were made between the formative (S1) and final submission based on professor feedback and internal review:

✓ Slide-based navigation system

The single-scroll dashboard was restructured into 8 full-screen slides, each dedicated to one analytical question in Marcus's journey. This significantly improves readability and allows each chart to be displayed at a larger size.

✓ Larger and more readable chart titles

All slide titles were redesigned — increased to 38px bold with key words highlighted in blue, making the analytical question immediately clear without reading the chart.

✓ Interactive chapter menu

A collapsible side panel and a cover page with clickable chapter cards allow navigation to any slide directly, not just sequentially with Prev/Next buttons.

✓ Airport map: improved readability

Added a ranked sidebar showing the top delayed and best-performing airports with exact delay figures. Fixed the Great Lakes white artifact by setting lakecolor to match the dark background. Added a contextual insight strip above the map.

✓ Marcus quote redesigned as speech bubble

The Marcus narrative quote was restyled from a small italic line to a proper speech bubble with avatar, name, role and location — making the character more prominent and the storytelling clearer.

✓ New KPI slide with Marcus verdict

Slide 01 now includes a Marcus's Verdict box (Delta · FRI 84 · Tuesday morning) and a live data pulse ticker that cycles through key statistics, making the opening slide more impactful and less static.

✓ Donut chart with breakdown sidebar

The cause donut chart was redesigned to include a right-hand sidebar with each category's percentage in large type and a '67% controllable' summary box — replacing the previous small legend.

Data sources

This memo was written by the Datafonos team as part of the Big Data Visualization module, Business Analytics Degree. The dashboard is implemented in Python using Plotly Dash and is accessible at <http://127.0.0.1:8050> when running locally. Navigation between the 8 analytical slides follows Marcus Reid's data journey from problem discovery to final decision.